

Tidal locking phenomenon — Solution

X23 (2023) — English translation

A1

0.70 pts

Obtain an exact expression for the angular velocity of orbital rotation ω . Express your answer in terms of G, m, M and L .

Let $\vec{e}_1$ be a unit vector drawn towards the smaller body from the center of the larger one. Acceleration of bodies due to gravity:

$$\vec{a}_m = -\frac{GM}{L^2}\vec{e}_1 \quad \vec{a}_M = \frac{Gm}{L^2}\vec{e}_1; \quad \vec{a}_M - \vec{a}_m = \frac{G(M+m)}{L^3}\vec{e}_1.$$

On the other hand:

$$\vec{a}_M - \vec{a}_m = \omega^2 L \vec{e}_1,$$

from where finally:

Answer: \textbf{Answer:}

$$\omega = \sqrt{\frac{G(M+m)}{L^3}}$$

A2

0.20 pts

Obtain an exact expression for the potential difference of the gravitational field of a small body $\Delta\varphi_{gp} = \varphi_{gpP} - \varphi_{gpO}$ at points P and O . Express your answer using G, m, L, θ and $r(\theta)$.

The field potential of a small body at a point whose distance from it is L_m is equal to:

$$\varphi_{gp} = -\frac{Gm}{L_m}$$

Let us find distance от малого тела до точки A по теореме косинусов:

$$L_m^2 = L^2 + r^2(\theta) - 2Lr(\theta)\cos\theta$$

Finally we have:

Answer: \textbf{Answer:}

$$\Delta\varphi_{gp} = Gm \left(\frac{1}{L} - \frac{1}{\sqrt{L^2 + r^2 - 2Lr\cos\theta}} \right)$$

A3

0.50 pts

Obtain an exact expression for the potential difference of inertial forces $\Delta\varphi_{ин} = \varphi_{инP} - \varphi_{инO}$ at points P and O . Express your answer using G, m, M, L, θ and $r(\theta)$.

Since the fluid is stationary, and the larger body rotates with a constant angular velocity, the fields of inertial forces $-\Delta m \vec{a}_0$ and $\Delta m \omega^2 \vec{r}_\perp$, acting on a particle of mass Δm , contribute to the potential difference. For the potential difference between points A and

O we have:

$$\varphi_{\text{иHA}} - \varphi_{\text{иHO}} = - \int_O^A (\omega^2 \vec{r}_\perp - \vec{a}_0) \cdot d\vec{r}$$

Taking into account that $\vec{a}_0 = Gm\vec{e}_1/L^2$, а $\vec{r}_\perp = \vec{r}$:

$$\varphi_{\text{иHA}} - \varphi_{\text{иHO}} = - \int_O^A \left(\omega^2 \vec{r} - \frac{Gm}{L^2} \vec{e}_1 \right) \cdot d\vec{r} = \frac{Gmr \cos \theta}{L^2} - \frac{\omega^2 r^2}{2}$$

Finally:

Answer: \textbf{Answer:}

$$\Delta\varphi_{\text{иH}} = \frac{Gmr \cos \theta}{L^2} - \frac{G(M+m)r^2}{2L^3}$$

A4

1.30 pts

Get addicted $h(\theta)$. Express your answer in terms of m, M, R, L and θ . Keep your answer as simple as possible. Qualitatively depict the shape of the surface in the section under consideration. In the same figure, depict the unperturbed shape of the surface. Note: Use the following approximation:

$$\frac{1}{\sqrt{1+a^2-2a\cos\theta}} \approx 1 + a\cos\theta + \frac{a^2(3\cos^2\theta-1)}{2}$$

Note: Use the following approximation:

$$\frac{1}{\sqrt{1+a^2-2a\cos\theta}} \approx 1 + a\cos\theta + \frac{a^2(3\cos^2\theta-1)}{2}$$

The surface of the larger body at equilibrium is equipotential. Since the surface disturbances are small, we will assume that the potential of the gravitational field of the larger body on the surface is equal to:

$$\varphi_M = -\frac{GM}{r(\theta)}$$

Since surface эквандпотенциальна:

$$\varphi_A = Gm \left(\frac{1}{L} - \frac{1}{\sqrt{L^2 + r^2 - 2rL\cos\theta}} \right) + \frac{Gmr \cos \theta}{L^2} - \frac{G(M+m)r^2}{2L^3} - \frac{GM}{r(\theta)} + C_1 = C_2$$

where C_1 and C_2 - некоторые постоянные величины. Раскладывая expression под корнем:

$$\frac{1}{\sqrt{L^2 + r^2 - 2rL\cos\theta}} \approx \frac{1}{L} \left(1 + a\cos\theta + \frac{a^2(3\cos^2\theta-1)}{2} \right); \quad a = \frac{r}{L}.$$

Thus, we have:

$$-\frac{GM}{r} - \frac{G(M+3m\cos^2\theta)r^2}{2L^3} = \text{const}$$

Note that terms, пропорциональные $\cos\theta$ полностью сократились. From-за этого and потребовалось раскладывать potential малого тела до второго порядка. Next substituting $r = R + h$ and expanding до первого порядка potential большого тела. В potentiale малого тела можно ссу считать $r = R$, since он сам является малой величиной from-за условия $R \ll L$. We get

$$-\frac{GM}{R} + \frac{GMh}{R^2} - \frac{G(M+3m\cos^2\theta)R^2}{2L^3} = \text{const}.$$

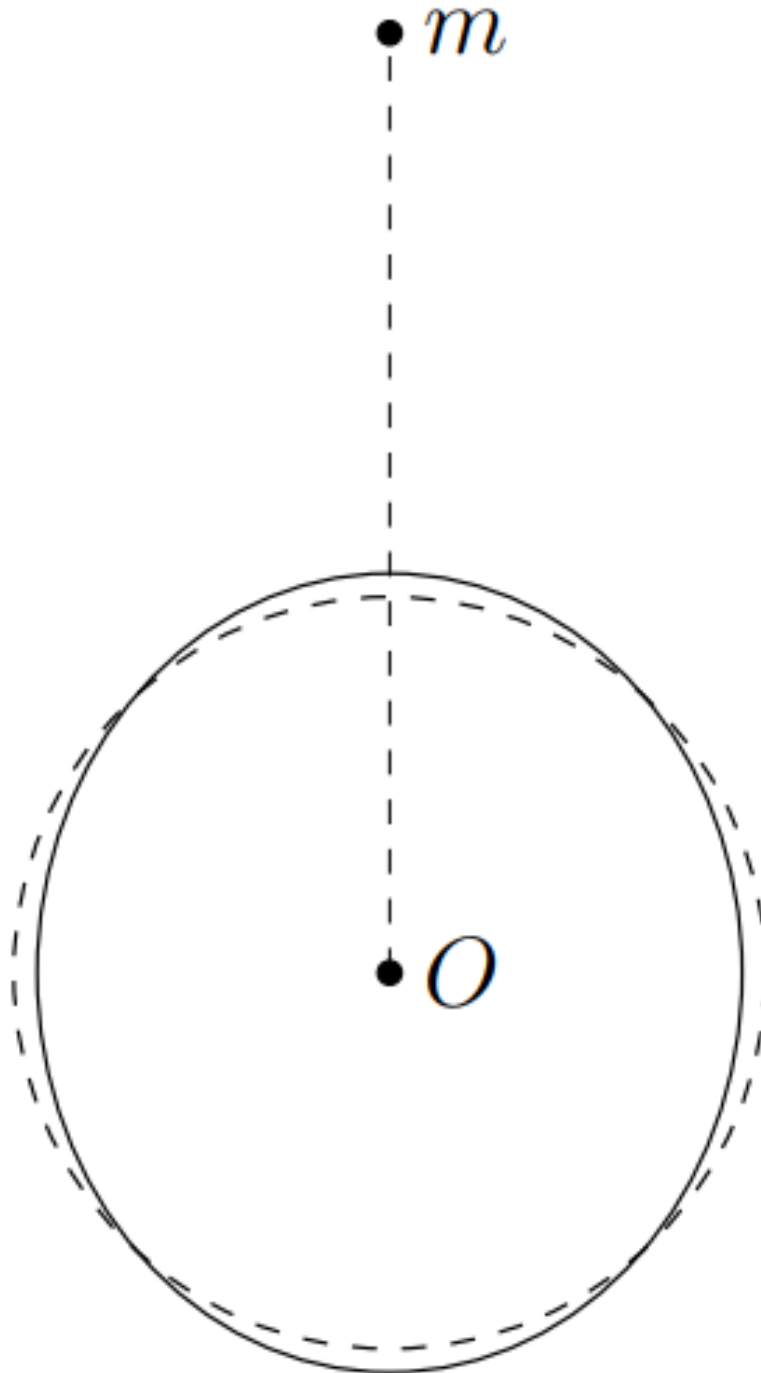
Оставим только переменные terms:

$$h \frac{GM}{R^2} - \frac{3GmR^2 \cos^2 \theta}{2L^3} = \text{const}$$

Since $h(\pi/2) = 0$, the last constant is zero. Then we have:

Answer: \textbf{Answer}:

$$h(\theta) = \frac{3mR^4 \cos^2 \theta}{2Md^3}$$



B1

0.50 pts

Show that the tangential component of the force $F_\tau(t, \theta)$ acting on a particle of mass Δm located on the surface of a larger body at an angle θ can be represented as:

$$F_\tau(t, \theta) = \Delta m \alpha \sin(2\omega_1 t - 2\theta)$$

Find α . Answer *вычитайте и делайте* via/through G, m, L and $r(\theta)$. For/Atмечание: воспользуйтесь результатами, полученными for/at решении parta A4.

Note: Use the results obtained when solving item A4.

In point A4 we obtained an approximate expression for the potential of inertial forces and the gravitational field of a smaller body:

$$\varphi_{\text{внеш}} = -\frac{G(M + 3m \cos^2 \theta)r^2}{2L^3}$$

Then for компоненты силы F_τ we get:

$$F_\tau = -\frac{\Delta m}{r} \frac{\partial \varphi_{\text{внеш}}}{\partial \theta} = -\frac{3Gm\Delta m r \sin 2\theta}{2L^3}$$

Отметим, что эта компонента силы возникает from-з-от однородности гравитационного поля меньшего тела. Since $\omega \neq \Omega$, в произвольный момент времени t parameter θ equals:

$$\theta = \theta_0 - \omega_1 t, \quad \omega_1 = \omega - \Omega.$$

Also учтем, что с нужной нам точностью $r \approx R$. Then finally we have:

$$F_\tau(t, \theta_0) = \frac{3Gm\Delta m R}{2L^3} \sin(2\omega_1 t - 2\theta_0)$$

Answer: \textbf{Answer:}

$$\alpha = \frac{3GmR}{2L^3}.$$

B2

0.80 pts

Show that the dependence $\theta(t, \theta_0)$ has the following form:

$$\theta(t) = \theta_0 + A \sin(2\omega_1 t - 2\theta_0 - \varphi_0)$$

Find A and φ_0 . Answer *вычитайте и делайте* via/through $G, m, L, \gamma, \omega_0, \omega, \Omega$ and θ_0 .

Let's substitute $F_\tau(t, \theta)$ into the equation of motion:

$$\Delta m r^2 (\ddot{\theta} + 2\gamma \dot{\theta} + \omega_0^2 (\theta - \theta_0)) = r F_\tau(t, \theta) = \Delta m r^2 \cdot \frac{3Gm}{2L^3} \sin(2\omega_1 t - 2\theta)$$

Since $\omega_0^2 \gg Gm/L^3$ - можно считать, что $F_\tau(t, \theta) \approx F_\tau(t, \theta_0)$. Then:

$$\ddot{\theta} + 2\gamma \dot{\theta} + \omega_0^2 (\theta - \theta_0) = \frac{3Gm}{2L^3} \sin(2\omega_1 t - 2\theta_0)$$

Solution уравнения будем искать в виде $\theta(t) = \theta_0 + \text{Re}(\hat{A} e^{2i(\omega - \Omega)t})$:

$$(\omega_0^2 - 4\omega_1^2 + 4i\omega_1\gamma) \hat{A} = \frac{3Gm}{2L^3} e^{-i(2\theta_0 + \pi/2)}$$

Let us represent $\hat{A}$ в exponent/indexной форме $\hat{A} = A e^{-i\varphi_0}$, where:

Answer: \textbf{Answer:}

$$A = \frac{3Gm}{2L^3 \sqrt{(\omega_0^2 - 4\omega_1^2)^2 + 16\gamma^2 \omega_1^2}}; \quad \varphi_0 = \arctan \frac{4\omega_1\gamma}{\omega_0^2 - 4\omega_1^2}.$$